Facial Pain Classification Using Deep Neural Networks for Clinical Decision Support

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Introduction

Accurate and real-time pain assessment is a critical challenge in healthcare, particularly for patients who are unable to communicate verbally, such as infants, post-operative patients, and individuals with neurological or cognitive impairments. Traditional pain assessment methods rely heavily on subjective observation, which can lead to inconsistent evaluations.

This project aims to develop a deep learning-based system that automatically detects and classifies pain intensity from facial images. The proposed system provides a non-invasive, objective, and scalable solution to support clinical decision-making and improve patient care.

Problem Statement

The objective of this project is to design a multi-class image classification model that categorizes facial images into four pain intensity levels:

- No Pain
- Mild Pain
- Moderate Pain
- Severe Pain

The model should achieve strong predictive performance, generalize well to unseen data, and support near real-time inference.

Dataset

This project utilizes the Balanced RGB Facial Expression dataset (7,575 images) from Kaggle. The dataset contains diverse facial images with balanced class distributions and sufficient variability in expressions.

Although the dataset is originally designed for facial expression recognition, it is used in this project as a proxy for pain-related facial patterns. Prior research suggests that certain facial expressions associated with emotions overlap with pain-related features, making this dataset suitable for exploratory modeling.

The dataset will be downloaded, organized into training, validation, and testing sets, and integrated into the preprocessing pipeline.

Methodology

Data Preprocessing

The dataset will undergo standard preprocessing steps, which may include resizing images, normalization, and data augmentation techniques such as flipping, rotation, and zooming. These steps may be adjusted and refined during experimentation to improve model performance.

Model Development

A baseline Convolutional Neural Network (CNN) will be implemented to establish initial performance. The CNN will include convolutional layers, pooling layers, and fully connected layers with appropriate regularization techniques such as dropout.

In addition to the baseline model, more advanced approaches will be explored, including transfer learning using pre-trained architectures such as ResNet50 and MobileNetV2. These models leverage learned visual features from large-scale datasets and can improve performance on limited data.

The final classification layer will be adapted to predict the four pain intensity levels, and selected models may be fine-tuned to improve performance. Other deep learning models may also be evaluated to compare results and identify the most effective approach. Model performance will be compared across different architectures to select the best-performing approach.

Training Strategy

Training configurations such as batch size, number of epochs, and learning rate may be adjusted during experimentation.

- Loss function: Categorical Cross-Entropy
- Optimizer: Adam
- Batch size: 32
- Epochs: 20–30 (with early stopping)

Regularization techniques:

- Dropout layers
- EarlyStopping (to prevent overfitting)
- ModelCheckpoint (to save best model)

Evaluation Metrics

The model will be evaluated using:

- Accuracy
- Precision
- Recall
- F1-score (macro)
- Confusion Matrix

These metrics ensure balanced evaluation across all classes, especially important for multi-class classification problems.

Model Interpretability

To improve model transparency, Grad-CAM will be used as an optional visualization technique to highlight important regions of the face that influence predictions. This helps provide intuition about the model's behavior, although the primary focus of the project remains on classification performance.

Deployment

The trained model will be deployed as a simple web application using Streamlit for the user interface. A lightweight backend using Flask (or direct integration within Streamlit) will be used for model inference. The application will be hosted on Hugging Face Spaces to enable easy access.

Users will be able to upload facial images and receive predicted pain intensity levels. If implemented, Grad-CAM visualizations may also be displayed to provide additional insight into model predictions.

Expected Outcome

The expected outcome is a complete deep learning pipeline capable of classifying pain intensity from facial images with competitive predictive performance across different model architectures. The project will demonstrate:

- The effectiveness of a baseline CNN model and the performance gains achieved using transfer learning approaches
- Performance improvements using pre-trained models
- A functional web-based application for real-time prediction
- (Optional) Visualization of model attention using Grad-CAM

This project highlights the potential of deep learning in assistive healthcare, particularly for patients who cannot communicate their pain effectively.